VIETNAM NATIONAL UNIVERSITY, HO CHI MINH CITY
UNIVERSITY OF TECHNOLOGY
FACULTY OF COMPUTER SCIENCE AND ENGINEERING



MATHEMATICAL MODELING (CO2011)

Assignment

"Dynamics of Love"

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List of Symbols

$\mathbb{N}$ Set of natural numbers

$\mathbb{R}$ Set of real numbers

$\mathbb{R}^+$ Set of positive real numbers

List of Acronyms

ODE (First-Order) Ordinary Differential Equation

IVP Initial-Value Problem

LTE Local Truncation Error

DS Dynamical System

Fig. Figure

Tab. Table

Sys. System of Equations

Eq. Equation

e.g. For Example

i.e. That Is

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Member list & Workload

No.	Fullname	Student ID	Problems	% done
1	Trần Anh Khoa	2113773	- Exercise 1: 1.2 - Exercise 2 - Exercise 3: 3.2	100%
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3	Lê Nguyên Chương	2112945	- Exercise 1: 1.1 - Exercise 4 - LaTeX	100%
4	Bùi Tiến Dũng	2113055	- Exercise 1 - Exercise 3: 3.1 - Exercise 4	100%

Table 1: Member list & workload

1 Introduction

Hmm [1]



Figure 1: Red-black tree

```
import numpy as np

def incmatrix(genl1,genl2):
    m = len(genl1)
    n = len(genl2)
    M = None #to become the incidence matrix
    VT = np.zeros((n*m,1), int)  #dummy variable

    #compute the bitwise xor matrix
    M1 = bitxormatrix(genl1)
    M2 = np.triu(bitxormatrix(genl2),1)

    for i in range(m-1):
        for j in range(i+1, m):
            [r,c] = np.where(M2 == M1[i,j])
            for k in range(len(r)):
                VT[(i)*n + r[k]] = 1;
                VT[(i)*n + c[k]] = 1;
                VT[(j)*n + r[k]] = 1;
                VT[(j)*n + c[k]] = 1;

                if M is None:
                    M = np.copy(VT)
                else:
                    M = np.concatenate((M, VT), 1)

```

```
                VT = np.zeros((n*m,1), int)

    return M
```

Listing 1: Python example

2 Conclusion

References

[1] Donald E. Knuth. *The TEX Book*. Addison-Wesley Professional, 1986.